PROJECT 8 · EC4.402 INTRODUCTION TO UAV DESIGN · TEAM 4

Enclosed Ducted-Fan Quadrotor: Component List (BOM)

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CONFIGURATION	ALL-UP MASS	THRUST-TO-WEIGHT	HOVER TIME	PEAK CURRENT	OUTER BOX
5-inch, 6S	**1,825 g**	**3.10**	**5.7 / 4.8 min**	**160 A**	**353 × 353 mm**
Skids + cradle; 180 g payload	1,905 g with leg option	1.41 kgf per motor, usable	0 / +20% hover power (enclosure loss)	Pack 169 A · ESC 4 × 60 A	~100 mm tall; 4 printed quarters

1. Flight hardware

#	Item	Selection	Qty to get	On aircraft	Mass ea. (g)	Counted (g)	Basis	Notes
Propulsion and power								
1	**Motors**	Emax ECO II 2207 1700 KV (6S)	4 + 1 spare	4	33.4	133.6	Spec	Listed 1.98 kgf; usable ≈ 1.41 kgf (× 0.89 air density × 0.80 voltage sag). Alt: iFlight XING2 2207 1750 KV.
2	**Motor leads**	Allowance, listed mass excludes wires	—	4	4.0	18.4	Est.	Weigh motors with leads on arrival.
3	**Propellers, 3-blade**	5 × 4.3 tri-blade (HQProp / Gemfan)	2 sets + 2 spare sets	4	4.5	20.7	Est.	Measure true diameter vs 127 mm before printing ducts. Balance every prop.
4	**Propellers, 2-blade**	5 × 4–4.5 two-blade, same brand	2 sets	0	4.0	—	Est.	Thrust-stand comparison only (decision D3).
5	**ESC**	SpeedyBee BLS **60 A** 4-in-1 (F405 V4 stack), 6S, current sensor	1 + 1 spare	1	23.5	23.5	Spec	60 A per channel vs ~40 A per motor (≥ 1.3× margin at 5-inch, 7-inch and 4S). The 55 A version is being discontinued. Mass as the 55 A board; confirm on arrival. BLHeli_S has no ESC telemetry: flash Bluejay for RPM over bidirectional DShot (harmonic notch). Current-sensor scale 400; check it doesn't clip at 160 A. Alt: T-Motor F55A Pro II.
6	**Battery**	6S 2600 mAh, ≥ 65 C, XT90	2–3	1	400	400.0	Spec	160 A peak vs 2.6 × 65 = 169 A (75 C preferred if available). A 45 C pack needs MOT_BAT_CURR_MAX ≈ 110 A, which cuts thrust to ≈ 1.10 kgf per motor (T/W ≈ 2.4 with skids, below the 2.5 target). Must fit the 45 mm spine: confirm dimensions.
7	**Capacitor**	Low-ESR, ~1000 µF, ≥ 50 V	1 spare	1	—	in #26	—	Normally supplied with the stack; listed as a spare. On the ESC power pads; absorbs 6S voltage spikes.
8	**Battery lead**	XT90 pigtail, 12 AWG	1 + 1 spare	1	—	in #26	—	
9	**Silicone wire**	12 AWG battery; 16–18 AWG motor extensions	1 m each	—	—	in #26	—	
Avionics and radios								
10	**Flight controller**	SpeedyBee F405 V4 (ArduPilot target speedybeef4v4)	1 + 1 spare	1	10.5	10.5	Spec	Baro, SD logging, 6S sensing. ArduCopter firmware. Alt: Matek H743-SLIM V3.
11	**GPS + compass**	Matek M10Q-5883 (u-blox M10, SBAS)	1	1	8.0	8.0	Spec	Top deck, away from power leads, never under carbon.
12	**GPS mast + cable**	Short mast, PA-CF base	1	1	10.0	11.5	Est.	
13	**RC receiver + telemetry (air)**	Lab RC receiver and telemetry radio, 433 or 865–867 MHz	1 each	1	30	30.0	Allow.	Replace with the lab models' mass. 915 MHz not permitted in India.
14	**RC transmitter**	Lab set (safety-pilot override)	1	—	—	—	—	Ground equipment.
15	**Telemetry radio (ground)**	Lab set + Mission Planner laptop	1	—	—	—	—	Ground equipment.
16	**Buzzer**	Active buzzer (arming, lost-model alarm)	1	1	3.0	3.5	Est.	

#	Item	Selection	Qty to get	On aircraft	Mass ea. (g)	Counted (g)	Basis	Notes
Payload (180 g ≈ 10% of all-up mass, the value assumed throughout our sizing)								
17	**Camera**	RunCam Thumb (1080p60, 5 V)	1	1	9.8	9.8	Spec	In the 60 × 40 × 30 mm bay behind a window; records the mission.
18	**Ballast**	Fixed mass in the payload bay	1	1	170.2	170.2	Defined	Brings the payload to 180 g.

2. Structure and printed parts

Estimates carry +15% contingency

#	Item	Selection	Qty to get	On aircraft	Mass ea. (g)	Counted (g)	Basis	Notes
19	**Frame**	Carbon true-X, 4 mm plate; 97.5 mm motor offset (276 mm motor to motor)	1	1	80	92.0	Est.	Or arms cut from carbon plate in the lab.
20	**Duct rings**	**PETG** (PETG-CF if hardened nozzle): bellmouth, 130 mm bore, 1.5 mm tip gap, 146.2 mm OD	4 + test rings	4	50	230.0	Est.	Test rings at 1.0 / 1.5 / 2.0 mm gap. Print upright, ≥ 1.6 mm walls.
21	**Side panels**	**PLA**, 0.8 mm	1 set	1	135	155.3	Est.	Lightweight PLA if over budget.
22	**Grilles (intake + exhaust)**	**PLA** square grid, 1 mm bars, 12 mm pitch, 4 mm deep	8	8	17	156.4	Est.	11 mm finger-safe openings, ~16% blockage. 3 mm deep if over budget.
23	**Decks, ribs, quick-attach joints**	**PLA**, 4 enclosure quarters (~177 mm, fit 256 mm bed)	1 set	1	142	163.3	Est.	
24	**Spine and bay walls**	**PLA**, 45 mm cross spine	1	1	30	34.5	Est.	3.8 mm clearance to each duct.
25	**Payload bay door, hinge, latch**	**PLA**, flush when shut, camera window	1	1	30	34.5	Est.	
26	**Wiring, connectors, inserts, fasteners**	Items 7–9, 29–34 on the aircraft	—	1	80	92.0	Est.	May reach ~110 g once every fastener is counted; within the overall contingency. Weighed first.
Landing gear — primary: flush skids + ground cradle								
27	**Skid pads**	**TPU**, flush with the box bottom	4 + spares	4	6	27.6	Est.	No mechanism; nothing hangs below the box.
28	**Ground cradle**	Open-frame take-off / landing stand, ≥ 250 mm tall (plywood or PVC)	1	—	—	—	—	Ground equipment. Lets the exhaust escape (≥ 1.5 × duct OD); wide top for ~±1 m landing accuracy. ≥ 320 mm for the 7-inch fallback.
All-up mass, 5-inch, skids, 180 g payload						**1,825.2**	**Datasheet 585.4 · estimates 1,039.6 · allowance + ballast 200.2**	

3. Optional: retractable legs (fitted if Gate 1 gives ≥ 1.19 kgf per motor)

#	Item	Selection	Qty to get	On aircraft	Mass ea. (g)	Counted (g)	Basis	Notes
A1	**Leg servos**	MG90S micro servo, metal gear	2 + 1 spare	2	13.4	26.8	Spec	ArduPilot landing-gear output (SERVOn_FUNCTION = 29). End points short of the hard stops.
A2	**Servo supply**	UBEC 5 V, ≥ 3 A, 2–6S input	1	1	10	11.5	Est.	Servos only, common ground; a stall can't brown out the FC.
A3	**Legs**	Printed, 2 × ~240 mm with lateral feet, hard stops; **PA-CF** hinges	2 + spares	2	25	57.5	Est.	Clears the 219 mm exhaust rule (1.5 × duct OD) with margin. Hinges offset in height or sideways so both legs fold flush past each other. Mass to confirm at this length.
A4	**Servo mounts**	**PA-CF**	1 set	1	10	11.5	Est.	
A5	**Hinge pins, servo leads**	M3 shoulder bolts / 3 mm pins; 2 extension leads	2 + spares	—	—	—	—	
All-up mass with legs instead of skids (+107.3 − 27.6)						**1,904.9**	**T/W 2.97 · hover 5.4 / 4.5 min**	

4. 7-inch fallback set

Used if Gate 1 thrust is below target; ordering early avoids a 2–4 week wait

#	Item	Selection	Qty to get	On aircraft	Mass ea. (g)	Counted (g)	Basis	Notes
F1	**Motors**	Emax ECO II 2807 1300 KV (6S)	4 + 1 spare	4	47.6	190.4	Spec	Replaces item 1. 2.48 kgf measured for a 2807 1300 KV on the 7040-3 at 24 V ≈ 1.77 kgf usable.
F2	**Propellers, 3-blade**	Gemfan Flash 7040-3, 5 mm hub	2 sets + 1 spare set	4	7.9	31.6	Spec	Replaces item 3. Measure true diameter vs 177.8 mm before printing ducts.
F3	**Battery**	6S 3300 mAh, ≥ 60 C, XT90	1–2	1	520	520.0	Spec	Replaces item 6. ~148 A full throttle (4 × 36.8 A, measured with the 2.48 kgf test at 24 V) vs 3.3 × 60 = 198 A; ~37–42 A per motor is within the 60 A ESC. Needs the 55 mm spine.
F4	**Frame**	Long-arm carbon true-X, ~369 mm motor to motor (130.5 mm offset)	1	1	135	155.3	Est.	Replaces item 19. Or arms cut from 4 mm carbon plate.
F5	**Duct rings, 7-inch**	**2.3–2.5 mm tip gap** (arms flex ~1.3 mm at full thrust), bore ~182.5 mm, OD ~204 mm	4 + test rings	4	100	460.0	Est.	Keeps ~1 mm residual clearance as at 5-inch. Box ~477 mm; duct gap 57.4 mm vs 55 mm spine; cradle ≥ 320 mm still clears (needs ~306 mm).
F6	**Other printed parts, rescaled**	Box structure 543 g, grilles 299 g, skids 41 g; wiring 104 g	1 set	1	—	987.0	Est.	All printed parts ≈ 1,343 g (802 g at 5-inch). **7-inch build needs ~2× filament and print time; book printer slots accordingly.**
All-up mass, 7-inch, skids, 180 g payload						**≈ 2,629**		**T/W ≈ 2.7 · = F1–F6 + leads, avionics, payload as at 5-inch (18.4 + 87 + 180). ESC, FC, GPS, radios, camera unchanged**

5. Build materials and consumables

#	Item	Selection	Quantity
29	**PLA filament**	Enclosure quarters, panels, grilles, spine, bay door	1–2 kg
30	**PETG / PETG-CF filament**	Duct rings (CF needs a hardened nozzle)	0.5 kg
31	**PA-CF filament**	Leg hinges and mounts, GPS mast base; needs dryer + enclosed printer	0.25 kg
32	**TPU filament**	Skid pads	0.1 kg
33	**Heat-set inserts**	M3 brass	30–40
34	**Fasteners**	Quarter-turn or printed clips; M3 × 6–12, M2	4–8 + assorted
35	**Vibration grommets**	FC stack	1 set
36	**Battery straps**	Non-slip, ~20 × 250 mm	2
37	**Heat shrink, zip ties, Velcro, threadlocker**	Assorted; medium-strength threadlocker for motor and frame bolts	—

6. Support and test equipment

#	Item	Selection
38	**Thrust stand**	Lab stand (or 5 kg load cell + HX711 + printed mount); logs thrust, current, RPM
39	**LiPo charger**	6S balance, ≥ 100 W
40	**LiPo safe bag**	Charging and storage
41	**Battery checker**	Per-cell voltage, before every flight
42	**Prop balancer**	Magnetic
43	**Soldering station**	≥ 60 W, for XT90 and motor leads
44	**Stage-1 test feet**	Simple skids for the bare-frame flights
45	**3D printer**	Bambu P1S (shared, 256 mm bed)

4S alternative (only if 6S packs are unavailable; motors and battery must be ordered together): Emax ECO II 2207 **~2,400 KV** in place of 1700 KV (same size and mounting), 4S **~4,000 mAh ≥ 60 C** (~430 g) in place of 6S 2600, 10 AWG battery lead with a connector rated above 180 A (e.g. AS150; XT90 is not), the BLS 60 A ESC already listed (45 A per motor × 1.3 = 58.5 A), ArduPilot battery and motor voltage limits set for 4 cells. Designed for T/W 2.5: ≈ 1,855 g all-up, hover ≈ 5.7 / 4.8 min, peak ≈ 180 A (~45 A per motor). Props, ducts, frame, enclosure, FC, GPS, servos and camera are unchanged. The 1700 KV motors would give only ≈ 0.62 kgf per motor on 4S.

Mass basis: Spec = seller datasheet, counted as listed · Est. = team estimate, counted +15% · Allow. = placeholder until the lab model is known · Defined = set by the team. Every part is weighed on arrival or after printing and the estimate replaced. Enclosure loss is modelled as 20% more hover power.